

Geeson Wan

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SUMMARY

Computer Engineering student at Purdue University with experience in programming, embedded development, and open-source software. Passionate about embedded systems, computer architecture, and semiconductors.

EDUCATION

2025 - present Undergraduate Computer Engineering at **Purdue University**
2021 - 2025 West Lafayette Junior/Senior High School (Academic Honors Diploma)

SKILLS

Programming Languages	Python, Java, C, C++, Ruby, Lua, HTML/CSS, JavaScript, & $\text{\LaTeX}$
Embedded	Arduino IDE, bread-boarding, basic electronics, microcontrollers
Frameworks	Ruby on Rails, StimulusJS
Tools/System	Git, Vim, Bash, & Linux
Languages Spoken	English (Native), Mandarin Chinese (Native)

PROFESSIONAL EXPERIENCE

Internship at Sledgehammer Infosystems June 2023 - August 2023

- Developed features for a Ruby on Rails web application using MVC architecture.
- Developed new features and bug fixes using Git-based collaborative development workflows.
- Researched and integrated unfamiliar tools and libraries through technical documentation

PROJECTS

Tempo Clock | Arduino, Embedded <https://github.com/Gerseneck/tempo-clock>

- Designed and implemented an embedded chess clock with configurable time controls, non-blocking firmware, and a menu-driven LCD interface.
- Developed a modular firmware architecture using state machines for timing, input handling, display updates, advanced logging, and persistent settings.
- Implemented EEPROM-based storage for user preferences and game statistics, along with software debouncing and audio alerts.

Game Boy Emulator | C <https://github.com/IEEE-C-Society-25-26/Game-Boy-Project>

- Implemented CPU instruction decoding and register emulation in C
- Refactored emulator subsystem interfaces to improve modularity and debugging
- Authored architecture and debugging documentation for emulator components

GatherPack | Ruby on Rails, StimulusJS, PostgreSQL <https://gatherpack.com>

- Developer of a full-stack event management platform using Ruby on Rails and StimulusJS
- Designed event scheduling subsystem supporting large events and calendar synchronization

- Designed responsive UI components optimized for mobile devices
- Currently used by multiple FIRST Robotics Teams as their core event management infrastructure

GreatStudier | Python

<https://github.com/greatericontop/GreatStudier>

- Developed an open source flashcard study command line application with a peer
- Designed and developed functionality for web scraping the contents for a study sets from Quizlet
- Wrote documentation and instructions regarding the usage and installation of the application

Translator For Open Source Projects

<https://hosted.weblate.org/user/Gerseneck/>

- Contributed Simplified Chinese and Traditional Chinese translations for System76 COSMIC DE, Flathub, Ubuntu, Fotema, Gaphor, and other projects
- Coordinated translation updates and managed localization feedback for open-source contributors

OTHER EXPERIENCES/ACHIEVEMENTS

Purdue Alpha Lambda Delta Phi Eta Honors Society

March 2026 - Present

National Honors Society | Tutor

August 2024 - May 2025

- A prestigious U.S. high school organization recognizing students for excellence in scholarship, leadership, Service, and character.
- Tutored Honors Chemistry for underclassmen

FIRST Robotics Team 461

August 2024 - May 2025

- Created the new team website (<https://github.com/frc461/website-2024>)
- Collaborated with team leadership to design a new scouting and match-analysis system
- Maintained technical infrastructure and organized development resources for team operations
- Experimented with a new Oculus based vision and location tracking system using OrangePI
- Developed multiple software and hardware subsystems, integrating embedded hardware with software components.

West Lafayette Hack Club

August 2023 - May 2025

- Executive Officer
- Planned and led weekly meetings for 20+ club members
- Taught CS workshops at multiple Hackathons that influenced over 100+ middle school and high school students
- Developed and maintained the team website using modern web technologies, improving communication and resource accessibility (<https://wl.hackclub.com>)

Volunteering with Boys Organization for Student Services

August 2021 - May 2024

- Volunteered for several Food Finders, Toy Drives, and Concession Stands